

Once you
stop learning,
you start dying.

—Albert Einstein

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also better for the environment, as it helps cut down on greenhouse gas emissions.

Supercapacitors beat batteries

Hybrid supercapacitors mark a significant step forward in energy storage, offering enhanced efficiency and longevity compared to traditional lead-acid and lithium batteries, especially in daily energy cycling applications. Lead-acid batteries, while cost-effective, fall short in environments requiring frequent deep cycling, typically enduring only three to five hundred cycles before their performance declines. Furthermore, they are highly susceptible to damage from deep discharges, which can permanently reduce their capacity and lifespan.

Lithium batteries, although more durable with about 2500 cycles at full capacity, still may not meet

the needs of systems that demand daily charging and discharging. Their higher initial cost and ongoing maintenance expenses often render them uneconomical, particularly when used primarily for backup in UPS (uninterruptible power supply) systems rather than for daily cycling.

In contrast, hybrid supercapacitors are specifically engineered for frequent use, capable of full charge and discharge cycles without the degradation typical of other battery types. This resilience makes them ideal for applications like storing solar energy during the day and utilising it during peak evening hours. For example, in our residential setup in Hawaii, hybrid supercapacitors contribute to a net zero annual energy bill by managing the daily cycle of solar energy, thereby decreasing dependence on external

power sources. These capacitors maintain their charge with minimal self-discharge—less than 2 percent per month—and are not adversely affected by extended periods of inactivity.

Moreover, hybrid supercapacitors demonstrate exceptional operational efficiency, achieving system efficiencies up to 97-98 percent, even considering inverter losses. This is significantly higher than the 80-85 percent efficiency of lead-acid batteries and the 90 percent efficiency of lithium batteries, largely due to reduced heat generation during charging and discharging, which indicates minimal energy loss. The superior cycle life and efficiency of hybrid supercapacitors not only improve their cost-effectiveness over time but also promote environmental sustainability by reducing peak-time demand on non-renewable